DE–3395

Sub. Code
13

DISTANCE EDUCATION

B.Sc. (Computer Science) DEGREE EXAMINATION, MAY 2016.

DIGITAL COMPUTER FUNDAMENTALS

(2007 Onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

$(5 \times 20 = 100)$

1. (a) Explain number codes and character codes.

   (b) Convert the following numbers from the given base to the other bases mentioned.
   - (i) $111110\ 111_2$ to decimal
   - (ii) $AB6_{16}$ to Decimal
   - (iii) $15\ F_{16}$ to Binary
   - (iv) 1568_{10} to Octal.

2. Explain the rules for the following OPERATIONS with suitable example.
   - (a) Binary Addition
   - (b) Binary Subtraction
   - (c) Binary 2's complement addition
   - (d) Binary 2's complement subtraction.

3. (a) Simplify the following Boolean functions using four Variable maps.
   - (i) $F(A,B,C,D) = \sum(0, 2, 4, 5, 6, 7, 8, 10, 13, 15)$
   - (ii) $F(A,B,C,D) = \sum(3, 7, 11, 13, 14, 15)$.

(b) Simplify the following expressions in product -of-sums form.

(i) $A'C + B'D + A'CD + ABCD$

(ii) $x'z' + y'z' + yz' + xy$.

4. (a) Explain with neat sketch Master Slave JK flip-flop.

(b) Simplify the following :

(i) $x + x'y$

(ii) $xy + x'z + yz$.

5. (a) Explain half adder and full adder with neat sketch.

(b) Design a 4-bit binary adder/subtractor. Explain its functions.

6. (a) Describe the following with functions, block diagram, waveforms and truth table.

(i) 4 bit-binary ripple counter

(ii) 4 bit-binary up-down counter.

(b) Design and explain NOR gate latch and NAND gate latch using RS-flip flop.

7. (a) Explain shift registers.

(b) Explain binary counter.

8. Explain in detail the concepts of input-output units and computer memory.

DE–3396

Sub. Code
14

DISTANCE EDUCATION

B.Sc. (Computer Science) DEGREE EXAMINATION, MAY 2016.

DATA STRUCTURES USING C

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Discuss the formatted and unformatted output statements of C with example. (12)

   (b) Define the following terms :

   (i) Constant

   (ii) Variable.

   Also list out rules for naming a variable. (8)

2. (a) What is the need for function? Discuss the function declaration and accessing method of a function. (12)

   (b) Define Array. Explain the multidimensional array with an example. (8)

3. (a) Explain method of declaring, initialization and accessing a variable through pointers. (10)

   (b) Write a 'C' program for arranging n numbers in ascending order. (10)

4. (a) Describe the operations of stack and implement stack using an array in a program. (10)

   (b) Write a C program to find nC_r value with recursive factorial function using the formula $nC_r = \frac{n!}{(r!*(n-r)!)}$. (10)

5. (a) What is Linked List? Explain the working of singly linked list with example. (12)

   (b) Discuss the concept of Binary trees in detail. (8)

6. (a) What is Queue? Explain the implementation of queue with array. (12)

   (b) Explain the following statements with examples : (8)

      (i) break

      (ii) continue

      (iii) goto

7. (a) Explain the structure concept in 'C' with simple example program. (8)

   (b) Write a note on various branching statement with examples. (12)

8. (a) Explain how elements of an array can be accessed using pointers. (10)

   (b) Write a 'C' program for multiplying two matrices. (10)

DE–3397

Sub. Code
15

DISTANCE EDUCATION

B.Sc. (Computer Science) DEGREE EXAMINATION, MAY 2016.

DISCRETE MATHEMATICS

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Construct truth table for $(P \vee Q) \vee \neg P$.

   (b) Prove $(P \to Q) \Leftrightarrow (\neg P \vee Q)$.

   (c) Explain about conditional and bi-conditional statements.

   (d) Symbolize the expression "All the world loves a lover". (5 + 5 + 5 + 5)

2. (a) What are set operations? Explain with examples.

   (b) Show that for any two sets A and B

   $A - (A \cap B) = A - B$. (10 + 10)

3. (a) Let $X = \{1, 2, ..., 7\}$ and

   $R = \{< x, y > | x - y \text{ is divisible by } 3\}$.

   Show that R is an equivalence relation.

   (b) Let R and S be two relations on a set of positive integers I.

   $R = \{< x, 2x > | x \in I\}$

   $S = \{< x, 7x > | x \in I\}$

   Find $R \circ S$, $R \circ R$, $R \circ R \circ R$ and $R \circ S \circ R$. (10 + 10)

4. (a) Write all possible functions from $x = \{1, 2\}$ to $y = \{a, b, c\}$ and classify them into one-to-one onto, neither one-to-one nor onto types of functions.

(b) If $f : A \to B$ and $B \to C$ are bijections, prove that $g \circ f : A \to C$ is also a bijections. (10 + 10)

5. (a) What do you mean algorithm? Explain the complexity of algorithms.

(b) Define and give examples for the following :

(i) Semi group

(ii) Monoid. (10 + 10)

6. (a) State and prove Lagrange's theorem.

(b) If $< G, * >$ is an abelian group, then for all $a, b \in G$. Show that $(a * b)^n = a^n * b^n$. (10 + 10)

7. (a) Find the left cosets of $\{[0], [3]\}$ in the group $< z_6, t_6 >$.

(b) Define the following with examples.

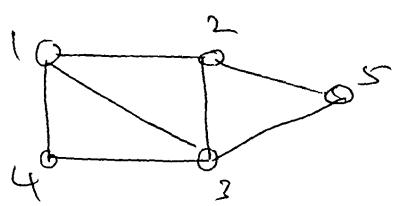
(i) Complete graph

(ii) Bipartite graph

(iii) Regular graph

(iv) Null graph. (10 + 10)

8. (a) Define : Adjacency matrix of the graph and find the adjacency matrix of the following graph.



(b) Explain the Konigsberg bridge problem. (10 + 10)

DE–9168

Sub. Code
16

DISTANCE EDUCATION

B.Sc. (Computer Science) DEGREE EXAMINATION, MAY 2016.

LAB I — C AND DATA STRUCTURES

(2007 onwards)

Time : Three hours Maximum : 100 marks

Examiner has to select ONE question and give it to the students by LOT system.

Answer both subdivisions (a) and (b) .

1. (a) Write a C program to find the area of a circle.

   (b) Write a C program to implement push and pop operations on stack.

---------- Cut here ----------

2. (a) Write a C program to calculate the compound interest.

   (b) Write a C program to implement insert and delete operations on Queue using array concept.

---------- Cut here ----------

3. (a) Write a C program to input a temperature in Celsius and find the corresponding temperature in Fahrenheit. Use the formula f = 9 / 5 * C + 32.

   (b) Write a C program to sort a set of elements using Selection Sort.

4. (a) Write a C program to concatenate two given strings and find length of the concatenated string.

   (b) Write a C program to sort a set of elements using Insertion Sort.

---------- Cut here ----------

5. (a) Write a C program to sort 10 numbers in ascending order.

   (b) Write a menu driven program in C to find an element using Linear and binary search methods.

DE–3398

Sub. Code
23

DISTANCE EDUCATION

B.Sc. (Computer Science)/B.Sc. (Computer Science) (Lateral)
DEGREE EXAMINATION, MAY 2016.

OBJECT ORIENTED PROGRAMMING AND C++

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Write a detailed note on various data types supported by C++ with example. (10)

   (b) Discuss the basic concepts of object oriented programming in detail. (10)

2. (a) Explain the concept of declaring and defining member function in a class with an example. (10)

   (b) Describe the working of Inline function with simple example program. (10)

3. (a) Illustrate the working of any four string manipulation functions with examples. (10)

   (b) Explain the use of constructor and destructor with examples. (10)

4. (a) Write a note on function overloading concept with an example. (10)

   (b) Write a C++ program to demonstrate the concept of binary operator overloading. (10)

5. (a) Explain the different types of inheritance in detail. (10)

   (b) Write a C++ program to prepare the pay bill using inheritance. (10)

6. (a) Explain the concept of virtual function with a sample program. (10)

   (b) Define a class complex. Using this class, overload '+' operator to add two complex numbers. (10)

7. (a) What is a pointer? Explain the use of *this* pointer with example program. (8)

   (b) With a suitable example show how a friend function can act as a bridge between two classes. (12)

8. (a) What is virtual base class? Explain with simple C++ program. (10)

   (b) Discuss the static and dynamic binding with examples. (10)

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Ws1

DE–3399

Sub. Code
24

DISTANCE EDUCATION

B.Sc. (Computer Science)/B.Sc (CS)(lateral). DEGREE EXAMINATION, MAY 2016.

SCIENTIFIC COMPUTING

(2007 Onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Solve the following system by gauss elimination method. (10)

$$2x + y + z = 10$$
$$3x + 2y + 3z = 18$$
$$x + 4y + 9z = 16.$$

(b) Using Gauss Jordan method, solve the following system of equations. (10)

$$2x - y + 3z = 8$$
$$-x + 2y + z = 4$$
$$3x + y - 4z = 0.$$

2. (a) Apply Gauss – Seidal method to solve the system of equations (10)

$$20x + y - 2z = 17$$
$$3x + 20y - z = -18$$
$$2x - 3y + 20z = 25.$$

(b) Construct Newton's forward interpolation polynomial for the following data: (10)

x	4	6	8	10
y	1	3	8	16

3. (a) Using Lagrange's interpolation formula compute $f(5)$: given $f(1) = 2, f(2) = 4, f(3) = 8, f(4) = 16$ and $f(7) = 128$ (10)

(b) Find $y'(x)$ given (10)

x	0	1	2	3	4
y	1	1	15	40	85

Hence find $y'(x)$ at x=0.5.

4. (a) Evaluate $\int_0^1 \frac{dx}{1+x}$ by using Trapezoidal rule, taking h=0.5 and h=0.25. (10)

(b) Calculate mean, median and mode from the following frequency distribution of marks at a test in statistics. (10)

Marks	5	10	15	20	25	30	40	45	50
No.of students	20	43	75	76	72	45	9	8	50

5. (a) Calculate the standard deviation from the following data: 14,22,9,15,20,17,12,11. (10)

(b) Fit a straight line trend to the data and estimate the expected profit for the year 1987. (10)

Year	1980	1981	1982	1983	1984	1985	1986
Profit (in lakhs)	60	72	75	65	80	85	95

6. (a) A husband and wife appears in an interview for two vacancies in the same post. The probability of husband's selection is (1/7) and that of wife's selection is (1/5). What is the probability that. (10)

(i) Both of them will be selected.

(ii) Only one of them will be selected.

(b) Two cards are drawn at random from a well suffled pack of 52 cards. What is the probability that. (10)

(i) Both are aces

(ii) Both are red

(iii) At least one is an ace.

7. (a) A petrol pump proprietor sells on an average Rs.80,000 worth of petrol on rainy days and an average of Rs.95,000 on clear days. Statistics from the meteorological department show that the probability is 0.76 for clear weather and 0.24 for rainy weather on coming Monday. Find the expected value of petrol sale on coming Monday. (10)

(b) A dealer in refrigerators estimates from his past experience the probabilities of his selling refrigerators in a day. These are as follows. (10)

No. of refrigerators sold in a day	0	1	2	3	4	5	6
probability	0.03	0.20	0.23	0.25	0.12	0.10	0.07

Find the number of refrigerators sold in a day.

8. (a) In an experiment on immunization of cattle from tuberculosis, the following results were obtained.(10)

	Affected	Not effected
Inoculated	12	26
not inoculated	16	6

Calculate χ^2 and discuss the effect of vaccine in Controlling susceptibility to tuberculosis.

(5% value of χ^2 for one degree of freedom=3.84)

(b) Explain the analysis of variance in two–way classification model. (10)

DE–3400

Sub. Code
25

DISTANCE EDUCATION

B.Sc. (Computer Science)/ B.Sc.(Computer Science) Lateral
DEGREE EXAMINATION,
MAY 2016.

APPLICATION PROGRAMS

(2007 Onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

($5 \times 20 = 100$)

1. (a) Explain about working with Windows Elements. (5)

   (b) Discuss about working with Multiple Windows. (5)

   (c) Discuss about the steps in Creating, Opening and Saving Files. (5)

   (d) What is a Menu? Explain. (5)

2. (a) Explain about creating Bullet and Numbered List in MS-WORD. (5)

   (b) What is a Header and Footer? Explain its creation in MS-WORD. (8)

   (c) Explain Mail Marge in MS-WORD. (7)

3. (a) Explain about Ranges in MS-EXCEL. (5)

   (b) Explain about Alignment and Number Formatting in MS-EXCEL. (8)

   (c) Discuss about Move, Copy, Find and Replace Data in MS-EXCEL. (7)

4. (a) Explain the Text Formatting and Alignment in MS-POWERPOINT. (8)

   (b) Discuss about Drawing on slides and Color Schemes. (7)

   (c) Explain Design Template and Auto Layout in MS-POWERPOINT. (5)

5. (a) Explain in detail about Database Creation in MS - ACCESS. (8)

   (b) How do you combine Excel data and Charts with Word ? Explain. (7)

   (c) Explain about Autoforms in MS-ACCESS. (5)

6. (a) Explain various Formatting Features in MS-WORD. (8)

   (b) Discuss about Tabs, Ruler, Margin and Paragraph Space in MS-WORD. (8)

   (c) Explain Templates and Wizards in MS-WORD. (4)

7. (a) Explain Insert, Delete and Resize Rows and Columns in MS-EXCEL. (6)

   (b) Discuss about Charts in MS-EXCEL. (8)

   (c) Explain Fill in Data, Autocomplete, Absolute Address in MS-EXCEL . (6)

8. (a) Explain in detail about Queries in MS-ACCESS. (15)

   (b) Discuss about Record Operations in MS-ACCESS. (5)

DE–9169

Sub. Code
26

DISTANCE EDUCATION

B.Sc. (CS)/B.Sc. (CS) (Lateral) DEGREE EXAMINATION, MAY 2016.

LAB II — C++ PROGRAMMING

(2007 onwards)

Time : Three hours Maximum : 100 marks

- Examiner has to select ONE question and give it to the student by LOT system.
- Each sub division carries 50 marks.

1. (a) Write a C++ program to reverse the sentence and find the given sentence is palindrome or not.

   (b) Create a C++ class for a stock item abstract data type. It should have the attributes of stock levels (an integer) and unit price (a float). Define the methods to return the values of these two attributes and to set them using parameters. Add two more methods to allow stock receipts and issues updating the stock level as appropriate. Write a menu driven C++ program to solve the problem.

---------- Cut here ----------

2. (a) Wire a menu driven C++ program to add and subtract given two matrices of order m × n defined in class. Using operator overloading.

   (b) Write a C++ program which will accept a string of 10 characters in length from the keyboard and count the occurrences of each of the five vowels in the string. The o/p should be in a (tapped) format similar to this example.

A	E	I	O	U
0	1	0	0	1

3. (a) Assumes you want to generate a table of multiples of any given number. Write a program that allows the user to enter the number, and then generates the table, formatting it into ten columns and 20 lines.

(b) Create a class called employee that contains a name and an employee number. Include a member function called getdata( ) to get data from the user, another function called putdata( ) to display the data. Write a main( ) program to exercise this class. It should create an array of type employee and then invite the user to input data for n employees.

---------- Cut here ----------

4. (a) Using dynamic constructors, write a C++ program to concatenate two given strings.

(b) Create a class INT to overload all five integer arithmetic operators (+, –, *, / and %) so that they operate on objects of type INT. If the result of any such arithmetic operation exceeds the normal range of int's - from –32, 768 to 32, 767 - have the operator print a warning and terminate the program. Write a C++ program to test this class.

---------- Cut here ----------

5. (a) Using pointers create a class and Write a C++ program to get the n names and display them in sorted order.

(b) Create a equivalent of four function calculator. The program should request the user to enter a number, an operator, and another number. It should then carry out the specified arithmetical operations: adding, multiplying, subtracting and dividing the two numbers. Finally it should display the result. When it finishes the calculation) the program should ask if the user wants to do another calculation.

6. (a) Create a class employee contains a Employee number, Employee name and address. Write a menu driven C++ program to get the 'n' number of employee details and display all details in employee namewise in sorted order.

(b) Write a C++ program that reads a group of numbers from the user and places them in array of type float. Once the numbers are sorted in the array, the program should average them and print the result. Use pointer notation whenever possible.

DE–9170

Sub. Code
27

DISTANCE EDUCATION

B.Sc. (CS)/B.Sc. (CS) (Lateral) (Computer Science) DEGREE EXAMINATION, MAY 2016.

LAB III — APPLICATION PROGRAMS

(2007 onwards)

Time : Three hours Maximum : 100 marks

Examiner has to select TWO questions ONE from Section A and another ONE from Section B.

SECTION A — (50 marks)

1. Prepare a news report using two columns, insert a picture in the first column and make the text flow around it.

---------- Cut here ----------

2. Create a table of student data that contains REGNO, NAME, ENGLISH, TAMIL, and MATHS marks. Add a new column named TOTAL and find the row total for each student.

   Add two rows named TOTAL, AVERAGE. Find the total and average values for each subject mark. Convert the table to text.

---------- Cut here ----------

3. Create a form letter that informs the customer about the date of maturity of a deposit amount in a commercial bank and request the customer for renewal. Mailmerge it with an Access Database containing all the customer data. Prepare letters for customers whose due date falls in a specific range of dates.

4. Create a Worksheet in MS-EXCEL with the following columns:

Employee number, Employee Name, designation, Basic pay, Hra, Da, Lic, Pf, Grosspay, and Netpay.

(a) Type data for empno, empname, designation, Basicpay, lic and pf.

(b) Calculate

hra	=	20% of basic
da	=	30% of Basic
Grosspay	=	basic+hra+da
Netpay	=	Grosspay — (lic + pf)

(c) Draw the bar chart between emp name and Netpay

(d) Sort the designation column and employee column name at a time.

---------- Cut here ----------

5. Create Internal Marks assessment worksheet with the following columns:

REGNO, NAME, SEX, TEST1, TEST2, TEST3, TEST, AVERAGE, ATTENDED DAYS, ATTENDANCE %, BONUS MARK, INTERNAL MARK.

(a) Add data for 20 students (Tests marks are out of 30)

(b) Compute TEST AVERAGE as average mark of best two of three tests.

(c) Maximum number of working days is 50. Each student should secure at least 80% attendance.

(d) Provide 1 bonus mark for each 1% attendance above eligibility limit 80%

(e) Compute Internal mark as TEST AVERAGE + Bonus Mark

(f) Sort the data in alphabetical order of name.

(g) Filter data for male and female students alone, who have attendance % below 80.

SECTION B — (50 marks)

6. Create Inventory table with fields ITEMNO, NAME, QUANTITY ON HAND and REORDER LEVEL.

   (a) Create a form in custom format.

   (b) Create a query REORDER to show all the items

   (c) Create a macro that executes REORDER query automatically.

---------- Cut here ----------

7. Create employee table with the fields EMPNO, NAME, AGE, SEX, STREET, CITY, PIN and SALARY.

   (a) Add data for 20 employees

   (b) Write a query to display all the male employees whose salary is between 1000 and 5000 and living in city "CHENNAI".

   (c) Write a query to display all the female employees whose ages are in the range 50-60

   (d) Show all the record in the table for all the city 'TRICHY' by filtering.

8. Create STUDENT table with fields REGNO,I1, E1, I2, E2, I3, E3, I4, E4, I5, E5 (internal and external marks in five subjects) and table SUBJECT with fields SCODE1, SNAME1, SCODE2, SNAME2, SCODE3, SNAME3, SCODE4, SNAME4,SCODE5 (Subject code and subject names for five subjects). Create a report to print marksheets for the entire student in the following format.

ALAGAPPA UNIVERSITY, KARAIKUDI

STATEMENT OF MARKS

REGNO: 05315001 NAME: RAMANA S

SUBCODE	SUBNAME	INTERNAL	EXTERNAL	TOTAL	RESULT
101	SYSTEM ANALYSIS	20	40	60	PASS
102	DBMS	22	50	72	PASS
103	VISUAL PROGRAMMING	20	38	58	PASS
104	INTERNET PROGRAMMING	21	27	48	FAIL
105	COMPILER DESIGN	28	30	58	PASS
			TOTAL	248	

(a) Condition for passing a subject: A minimum of 35 marks in external and 50 marks in total.

(b) Grand total should not include total in failed subjects.

(c) Display FAILS in red color.

DE–3401

Sub. Code
31

DISTANCE EDUCATION

B.Sc. (CS)/B.Sc. (CS) (Lateral) DEGREE EXAMINATION, MAY 2016.

COMPUTER GRAPHICS

(2007 Onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

1. (a) List and explain the Interactive Input devices. (10)

   (b) Demonstrate Midpoint Circle generation algorithm with example. (10)

2. (a) Discuss in detail about Windowing Transformation. (10)

   (b) Explain Cohen-Sutherland line clipping algorithm in detail. (10)

3. (a) Describe the basic three dimensional transformations. (10)

   (b) Narrate three dimensional composite transformations. (10)

4. (a) Write a note on 3D Viewing. (10)

   (b) Discuss any one Hidden Surface Removal algorithm. (10)

5. (a) Explain the components of user interface in detail. (10)

   (b) Write a detailed note on command language. (10)

6. (a) Discuss about the applications of computer graphics. (10)

   (b) Summarize about point clipping. (10)

7. (a) Write a note on View port and Aspect ratio. (10)

   (b) Explain Bresenham's line drawing algorithm with suitable example. (10)

8. (a) Discuss the concept of parallel projection in 3D. (10)

   (b) How the rotation of an object about the pivot point is performed? Explain. (10)

DE–3402

Sub. Code
32

DISTANCE EDUCATION

B.Sc. (CS)/B.Sc. (CS) (Lateral) DEGREE EXAMINATION, MAY 2016.

VISUAL PROGRAMMING

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Discuss about the format of Windows.

   (b) Explain the functions of minimizing, maximizing and restoring windows with mouse and the keyboard.

2. (a) What is icon? Describe the functions of four icons on the desktop.

   (b) Explain, how to use windows common user interface components in an application.

3. (a) Describe the various control panel operations.

   (b) Write down the procedure steps for installing multimedia devices.

4. (a) What are the various control structures supported by Visual Basic? Explain any two of them.

   (b) Write a Visual Basic program using function procedure to find the power of a number.

5. (a) Explain the different types of procedures of Visual Basic.

   (b) What are the two types of arrays in Visual Basic? Explain.

6. (a) Explain any five string functions of Visual Basic.

   (b) Write a Visual Basic program to display the current date and time in a Form.

7. (a) Name and describe the three styles of combo box.

   (b) Describe the three controls for creating graphical applications.

8. (a) Explain the various steps to create an MDI Application.

   (b) Using the FlexGrid control Write a Visual Basic program that performs addition, subtraction, multiplication and division of numbers.

DE–3403

Sub. Code
33

DISTANCE EDUCATION

B.Sc. (CS)/B.Sc. (CS) (Lateral) DEGREE EXAMINATION, MAY 2016.

OPERATING SYSTEMS

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) Explain operating system concepts.

   (b) Explain the structure of operating system with neat sketch.

2. Discuss about the various CPU scheduling algorithms.

3. (a) Explain the various disk scheduling techniques with an example.

   (b) Describe interrupt handlers.

4. Define deadlock. Explain in detail deadlock detection and deadlock prevention.

5. (a) Explain about contiguous memory allocation.

   (b) Explain the basic concepts of segmentation.

6. (a) Explain the following page table structure:

      (i) Hierarchical page Table

      (ii) Hashed Page table.

   (b) Explain the various page replacement strategies.

7. (a) What are files and explain the access methods for files?

   (b) Write notes about the protection mechanisms provided for file servers.

8. Write short notes on the following:

   (a) Race condition

   (b) Device drivers

   (c) Virtual memory

   (d) Free space management.

DE–3404

Sub. Code
34

DISTANCE EDUCATION

B.Sc. (Computer Science)/B.Sc. (CS) (Lateral) DEGREE EXAMINATION, MAY 2016.

INTERNET PROGRAMMING

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

$(5 \times 20 = 100)$

All questions carry equal marks.

1. (a) What is a network topology? Explain the different network topologies.

   (b) What is TCP/IP? Explain the protocol stack of TCP/IP.

2. (a) How are addresses defined on the Internet? Explain.

   (b) State the types of URL and explain them.

3. (a) What is FTP and how does it work?

   (b) What is e-mail? Explain its working with the help of an example.

4. (a) What are the various services provided by an Internet? Explain.

   (b) Explain the various applications of E-Commerce.

5. (a) What is an object-oriented programming? Why is java known as platform independent language?

   (b) Write a Java program to find the sum of the digits of a given integer.

6. (a) List the basic data types used in java. Explain with suitable example.

   (b) What are the classifications of operators in Java? Explain with examples.

7. (a) Discuss how I/O operations are performed in Java.

   (b) Write a Java program to illustrate the method overloading.

8. (a) Describe the different stages in the life cycle of an applet.

   (b) Write an applet to draw the square inside a circle.

DE–3405

Sub. Code
35

DISTANCE EDUCATION

B.Sc. (CS)/B.Sc. (CS) (Lateral) DEGREE EXAMINATION, MAY 2016.

RDBMS

(2007 onwards)

Time : Three hours Maximum : 100 marks

Answer any FIVE questions.

All questions carry equal marks.

(5 × 20 = 100)

1. (a) What is a DBMS? Describe the advantages of DBMS?

   (b) Explain the manipulative capabilities of database scheme.

2. (a) Explain the concepts of relational model.

   (b) Explain in detail about data manipulation facilities.

3. Explain in detail with neat sketch of Oracle architecture.

4. Explain all the set operations in Oracle with suitable SQL query.

5. (a) Describe table definition? Give example.

   (b) Explain Oracle database objects.

6. (a) Explain ORACLE queries working with NULL values in detail. Give example.

   (b) Explain subqueries in ORACLE. Give example.

7. (a) Discuss about embedded SQL.

   (b) Describe the benefits of Oracle.

8. Write short notes on the following :

   (a) Schema and subschema

   (b) Relational model: Mapping operation

   (c) Distinct value in SQL

   (d) Oracle query using expressions.

DE–9171

Sub. Code
36

DISTANCE EDUCATION

B.Sc. (CS)/B.Sc. (CS) (Lateral) DEGREE EXAMINATION, MAY 2016.

LAB IV — VISUAL BASIC AND ORACLE

(2007 onwards)

Time : Three hours Maximum : 100 marks

Examiner has to select TWO questions. One from Visual Basic and another from Oracle. The questions are given to the student by lot system.

Each sub division carries 50 marks.

VISUAL BASIC

1. Write the test a VB program to compute the intervals between two dates in terms of day, months and years. Use option buttons, text boxes and command buttons.

---------- Cut here ----------

2. Write a VB program to display and calculate simple interest or compound interest using three command buttons namely Input, Calculate, Display.

---------- Cut here ----------

3. Write a VB program to do temperate conversion C to F and F to C. Use option buttons and input boxes.

4. Write and test a VB program to compute and print either sum of odd numbers or even numbers at the user's choice using Label, Text, Option buttons.

---------- Cut here ----------

5. Write and test a VB Program to print twenty address labels in a two-column format for the address.

THE REGISTRAR

ALAGAPPA UNIVERISTY

KARAIKUDI – 630003.

---------- Cut here ----------

6. Write and test a VB program to compute the surface area and volume of a cylinder given radius and height. Use option buttons and input boxes.

---------- Cut here ----------

7. Write a VB program to compute and print the product of the first N natural numbers. Use option button.

---------- Cut here ----------

8. Write a VB program that allows the user choice among four arithmetic operations (+, –, ∗ and /) with given two numbers. Use option buttons and input boxes.

ORACLE

9. Create table NUMB with the following structure :

Field Name	Type	Width	Decimal
NVM	Numeric	6	2
ABSNO	Numeric	6	2
SQRTNO	Numeric	6	2
ROUNDNO	Numeric	6	2
TRUNCNO	Numeric	6	2
SQRNO	Numeric	6	2
CUBENO	Numeric	6	2

(a) Add 5 records (Some numbers must be negative (or) decimal).

(b) Display all the records.

(c) Replace ABSNO with absolute value of given numbers.

(d) Replace SQRTNO with square root value of given numbers.

(e) Replace round No by using ROUND function.

(f) Replace TRUNCNO by using TRUNC function.

(g) Replace SQRNO with the square of given numbers.

(h) Replace CUBNO with CUBE of given numbers.

(i) Display all the records.

10. Create a table BIO-DATA with the following structure :

Field Name	Type	Width	Decimal
Name	Char	25	
Age	Numeric	2	
Sex	Char	1	
Degree	Char	8	
City	Char	20	

(a) Add 5 records (BIO-DATA of 5 employees).

(b) Display all records of MALE employees.

(c) Delete all B.Sc. degree holders and then display the table contents.

(d) Remove all employee records whose AGE is greater than 20.

(e) Display all records having AGE < 20 and CITY is TIRUNELVELY.

---------- Cut here ----------

11. Create table MARK with the following structure

Field name	Type	Width	Decimal
REGNO	Numeric	8	
NAME	Character	20	
TAMIL	Numeric	3	
ENGLISH	Numeric	3	
MATHS	Numeric	3	
BIOLOGY	Numeric	3	

(a) Add 5 Records.

(b) Show data in fields REGNO, NAME, TAMIL and BIOLOGY alone on the screen.

(c) Show data in fields REGNO, BIOLOGY and MATHS alone on the screen.

(d) Show data in fields NAME and REGNO alone on the screen.

(e) Show all fields on the Screen.

12. Create table 'SALES' with the following structure :

Field name	Type	Width	Decimal
SNO	Numeric	5	
SNAME	Character	25	
SAMT	Numeric	8	2
COMM	Numeric	6	2

(a) Add 5 records.

(b) Display the salesman details whose sales amount is maximum.

(c) Replace the 'COMM' with '0' if sales amount is less than 1000; 10% of sales amount if sales amount is above 1000 and below 10,000; 15% of sales amount if sales amount is equal to or above 10000.

(d) Display all records.

(e) Display the records having commission > 1000.

---------- Cut here ----------

13. Create table Bank with following structure :

Field name	Type	Width	Decimal
ACCOUNT NO	Numeric	10	
CNAME	Character	25	
TYPE	Character	1	(D-Deposit
DATE	Date		W-Withdraw)
AMT	Numeric	5	
BALANCE	Numeric	8	2

(a) Insert 5 records.

(b) Display all transactions with Type is 'D'.

(c) Display all transactions belonging to the customer with the account number '1512'.

(d) Display all the transactions on the date 10-Dec-2008.

(e) List all details whose balance is above 5000 on 15th March 2009.

14. Create table ADDRESS with the following structure :

Field name	Type	Width
NAME	Character	25
STREET	Character	30
CITY	Character	20
PINCODE	Numeric	6

(a) Add 5 Records.

(b) Display the structure of the table.

(c) Add the field DISTRICT.

(d) Fill all DISTRICT with KARAIKUDI.

(e) Change the pincode to 630003 where the city is Karaikudi with pincode as 630003.

---------- Cut here ----------

15. Create a VENDOR table with the following details :

Field name	Type	Width
VENDOR NAME	Varchar	20
VENDOR NO	Varchar	10
ADDRESS	Varchar	10
CITY	Varchar	15
STATE	Varchar	15
PRODUCT NAME	Varchar	15

(a) Add 8 Records.

(b) Change the width of the address field to 30.

(c) Display VENDOR NO, VENDOR NAME, ADDRESS, CITY for Tamilnadu State.

(d) Display the Vendor details who supply the Laptop.

(e) Display the distinct VENDOR NAME.

DE–9172

Sub. Code
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DISTANCE EDUCATION

B.Sc. (CS)/B.Sc. (CS)/(Lateral) DEGREE EXAMINATION, MAY 2016.

Lab V – JAVA PROGRAMMING

(2007 Onwards)

Time : Three hours Maximum : 100 marks

ONE question should be selected by the examiner for each candidate and given to them by lot system.

Each sub division carries 50 marks.

1. (a) Write a JAVA program for sorting a given list of names in ascending order using Command line arguments.

   (b) Write a JAVA program to count number of words & Characters in a text.

---------- Cut here ----------

2. (a) Write a JAVA program to multiply two given matrices.

   (b) Write a JAVA program to implement Producer/ Consumer problem using synchronization.

3. (a) Write a JAVA program which illustrates the implementation of abstract class.

(b) Write a JAVA program to create packages.

---------- Cut here ----------

4. (a) Write a JAVA program for illustrating Overloading and overloading methods.

(b) Write a JAVA program to write Applets to draw the various polygons.

---------- Cut here ----------

5. (a) Write a JAVA program to multiply two given matrices.

(b) Write a JAVA program to create and manipulate Labels, Lists, Text Fields, Text Areas and Panels.

---------- Cut here ----------

6. (a) Write a JAVA program to create multiple Threads.

(b) Write a JAVA program to illustrate the use of files and Streams.